

Markovian and multi-curve friendly parametrisation of a HJM model used in valuation adjustment of interest rate derivatives.

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Motivation

XVA NEEDS A JOINT, SIMULABLE CURVE MODEL

- Valuation adjustments (CVA, DVA, FVA and related XVA terms) on interest-rate derivative portfolios require simulating the entire curve system, discounting and projection curves jointly, over long horizons and across collateralised and uncollateralised netting sets.
- The Heath–Jarrow–Morton framework prices consistently by construction, but a generic HJM volatility makes the forward-curve state non-Markovian, which is computationally prohibitive in an XVA engine.
- The paper constructs a parametrisation that is simultaneously Markovian and consistent with the post-crisis multi-curve setting.

The HJM framework and the Markovian restriction

MODEL CLASS

Under the risk-neutral measure the instantaneous forward curve follows

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dW_t, \quad \alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, s) ds,$$

the drift pinned down by no-arbitrage. A separable volatility structure, $\sigma(t, T) = g(t) h(T)$, collapses the curve dynamics onto a finite-dimensional Markovian state.

- The paper works out a parametrisation in this class rich enough to fit the observed swap and OIS curve data while keeping the state vector small.

Multi-curve consistency

DISCOUNTING AND PROJECTION AFTER THE CRISIS

- Post-2008 markets separate the discounting curve (OIS, collateral-rate based) from tenor-specific projection curves, with non-negligible basis between them.
- The parametrisation is built to be multi-curve friendly: the discount and projection layers are modelled consistently, so collateralised and uncollateralised trades are valued within one engine.
- The result is a state-space small enough for the nested simulations of XVA while preserving the fit to the swap-curve and OIS-curve data.

| Implications

FOR XVA PRACTICE

- Dimensionality reduction is obtained by parametric structure instead of ad-hoc factor truncation, so the no-arbitrage consistency of HJM is retained inside the XVA engine.
- The framework covers collateralised and uncollateralised interest-rate derivatives in one model, which is precisely the portfolio mix on which valuation adjustments are computed.
- The construction predates and complements the later LLM yield-curve programme: the same discipline of explicit parametrisation over convention.

Thank you.

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